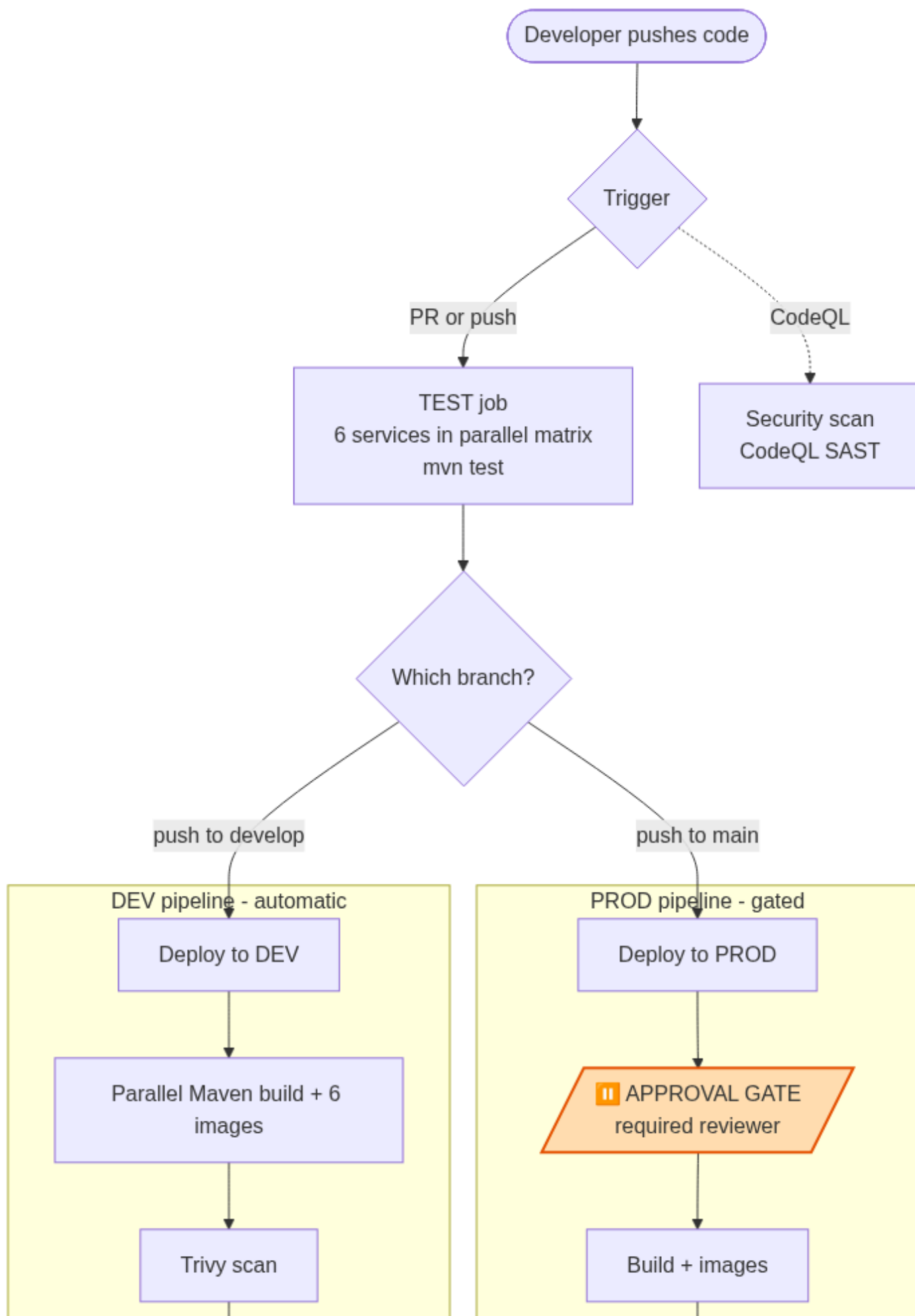
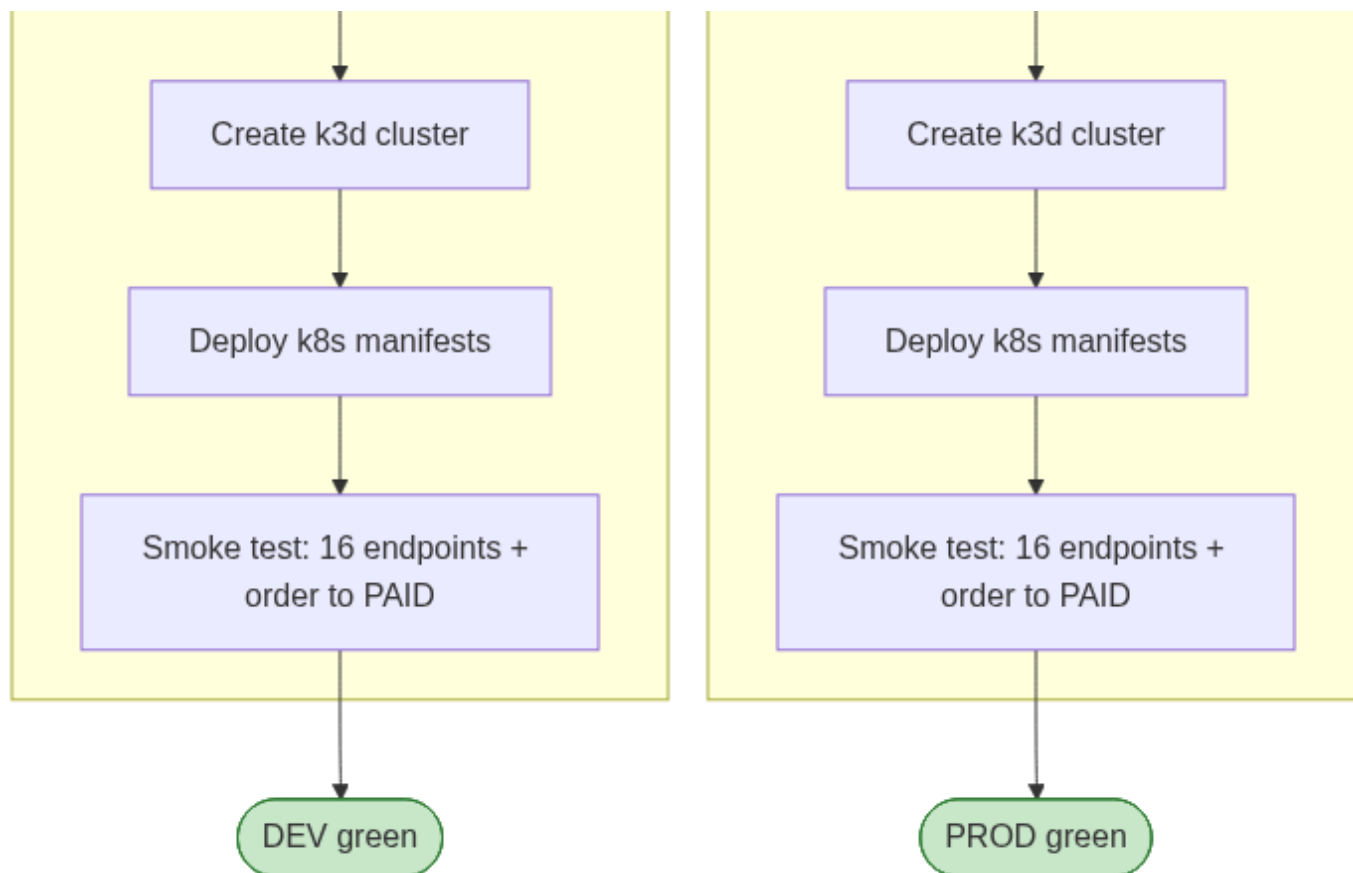


CI/CD Pipeline

Overview

Every code change goes through automated testing, security scanning, image building, and deployment before it ever runs in "production."





Job 1: Tests (runs on every PR and push)

File: `.github/workflows/ci-cd.yml` → `test` job

```
services:
  postgres: postgres:15      # Real DB for integration tests
  redis: redis:7-alpine      # Real Redis for cache tests
```

Why real services in CI? Mocks lie. We run tests against actual Postgres and Redis so that if the DB schema is wrong or a query fails, CI catches it — not production. **JaCoCo coverage gate:**

```
mvn test
→ generates: target/site/jacoco/index.html
→ fails if line coverage < 50%
→ excludes: dto/, model/, config/, exception/ (boilerplate)
```

CodeQL (SAST):

- Scans for SQL injection, XSS, insecure deserialization
- `continue-on-error: true` — private repos don't get Advanced Security for free; scan still runs, results go to Security tab

Job 2: Build + Push + Deploy (develop branch)

Triggered only when code is **pushed to** `develop` (not on PRs).

Step 1 — Start Floci

```
docker compose up -d floci
# Wait for: curl http://localhost:4566/_floci/health
```

Floci is the AWS simulator. ECR, SES, ALB APIs all respond at `localhost:4566`.

Step 2 — Configure AWS (Floci-safe)

```
# We do NOT use aws-actions/configure-aws-credentials
# (that action calls real AWS STS - rejects fake test/test credentials)
echo "AWS_ACCESS_KEY_ID=test" >> $GITHUB_ENV
echo "AWS_SECRET_ACCESS_KEY=test" >> $GITHUB_ENV
```

Step 3 — Build Images

```
for svc in user-service product-service ...; do
  cd $svc && mvn package -DskipTests -q && cd ..
  # Tag for Floci ECR
  docker build -t localhost:5100/ecommerce/$svc:$GIT_SHA .
  # Tag for docker-compose (image name = ecommerce-$svc)
  docker tag localhost:5100/ecommerce/$svc:latest ecommerce-$svc:latest
done
```

Images are tagged with the **git commit SHA** (`abc1234`).

This means every image is traceable back to the exact commit that produced it.

Step 4 — Trivy Security Scan

```
uses: aquasecurity/trivy-action@master
with:
  image-ref: localhost:5100/ecommerce/user-service:${{ steps.meta.outputs.version }}
  severity: CRITICAL,HIGH
  ignore-unfixed: true
  cache: true # avoids re-downloading 100MB vulnerability DB each run
```

Scans one representative image (all 6 share the same `eclipse-temurin:21-jre-alpine` base).

`exit-code: 0` — scan reports findings but doesn't fail the build (informational).

Step 5 — Push to Floci ECR

```
aws ecr get-login-password --endpoint-url http://localhost:4566 | \
  docker login --username AWS --password-stdin localhost:5100

docker push localhost:5100/ecommerce/user-service:abc1234
docker push localhost:5100/ecommerce/user-service:latest
```

Step 6 — Smoke Test via docker-compose

```
# Start everything (no-build = uses images we just built)
docker compose up -d --no-build

# Wait for ALL 3 services to be ready (not just one)
for i in $(seq 1 120); do
  P=$(curl -s -o /dev/null -w "%{http_code}" http://localhost:8000/api/products)
  U=$(curl -s -o /dev/null -w "%{http_code}" -X POST http://localhost:8000/api/users/login
  ...)
  O=$(curl -s -o /dev/null -w "%{http_code}" http://localhost:8000/api/orders/user/1)
  if [ "$P" = "200" ] && [ "$U" != "503" ] && [ "$O" != "503" ]; then
    break
  fi
  sleep 5
done

# Run smoke tests
curl -f -X POST http://localhost:8000/api/users/register ...
curl -f http://localhost:8000/api/products
curl -f -X POST http://localhost:8000/api/products ...
```

Why docker-compose and not K8s in CI? K8s (Kind) takes 5-10 minutes to start on a 2-CPU GitHub runner. docker-compose with pre-built images is ready in ~90 seconds. K8s manifests are tested locally with Floci EKS — CI verifies app logic.

Job 3: Production Deployment (main branch, approval required)

Same steps as Job 2 with one difference: `environment: production`.

This pauses the job at deployment time and requires a designated reviewer to click **"Approve deployment"** in GitHub:

```
GitHub → Actions → (failed run) → Review deployments → Approve and deploy
```

To set this up:

1. GitHub → Settings → Environments → New environment → `production`

2. Add required reviewers (e.g., tech lead)
3. Set wait timer: 5 minutes (prevents accidental approvals)

Dependabot (Automatic Dependency Updates)

File: `.github/dependabot.yml`

Every week, Dependabot opens PRs for:

- Maven dependencies (Spring Boot, Kafka, Redis client, etc.)
- Docker base images (`eclipse-temurin:21-jre-alpine`)
- GitHub Actions versions (`actions/checkout@v4` , etc.)

Each Dependabot PR runs the full test suite automatically. You merge it if tests pass.

Branch Protection Rules (Set Up in GitHub)

For `main` :

```
Settings → Branches → main
☒ Require PR before merging
☒ Require status checks: test
☒ Require approvals: 1
☒ Dismiss stale reviews on new commits
☒ Restrict force pushes
```

For `develop` :

```
☒ Require PR before merging
☒ Require status checks: test
☐ Approvals: optional (or 1 for teams)
```

Full Developer Workflow Summary

Feature work:

```
git checkout -b feature/xxx develop
[make changes + mvn test locally]
git push → open PR to develop
→ CI: tests run (Job 1)
→ Team review
→ Merge to develop → CI: build + deploy to dev (Job 2)
```

Release:

```
Open PR: develop → main
→ CI: tests run (Job 1)
→ Tech lead approves PR
→ Merge to main → CI: build + PAUSE for approval (Job 3)
→ Reviewer approves deployment
→ Deploy to production
```

Adding a New Service to CI

1. Add service to `SERVICES` env var in `.github/workflows/ci-cd.yml`
2. Add service folder with `pom.xml` and at least one unit test
3. Add `image: ecommerce-:latest` to `docker-compose.yml`
4. Add ECR repo creation to `scripts/03-ecr.sh`
5. Add K8s Deployment + HPA to `k8s/services/deployments.yaml`